

Mengran Li

Postdoctoral Research Associate

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Summary

I develop new statistical methodology for causal inference in rare and extreme events, spanning causal discovery, treatment-effect estimation and climate event attribution. My research combines causal inference with extreme value theory to enable reliable decision-making in high-impact extreme regimes.

Academic Appointments

Smith School of Enterprise and the Environment, University of Oxford

Postdoctoral Research Associate

AI-enabled causal evaluation of nature conservation

Oxford, UK

2026 – Present

Education

University of Glasgow

Doctor of Philosophy in Statistics

Thesis: *Causal Inference in the Tails: Extreme Event Attribution, Treatment Effects, and Structural Discovery*

Glasgow, UK

2022 – 2026

University of Glasgow

Master of Science in Statistics with distinction

Glasgow, UK

2021 – 2022

Southwestern University of Finance and Economics

Bachelor of Science in Statistics

Bachelor of Economics in Finance (Double Degree)

Chengdu, China

2015 – 2019

Publications

Peer-Reviewed Publications

* denotes corresponding author

1. **Li, M.***, Cuba, D., Hu, C., & Castro-Camilo, D. (2024). A wee exploration of techniques for risk assessments of extreme events. *Extremes*. doi:10.1007/s10687-024-00500-5

Manuscripts / Preprints

2. **Li, M.*** & Castro-Camilo, D. (2026). Causal Discovery in Multivariate Extremes via Tail Asymmetry. *arXiv:2604.21620*. Resubmission invited at *Journal of the American Statistical Association (Theory and Methods)*.
3. **Li, M.*** & Castro-Camilo, D. (2026). Tail-Calibrated Estimation of Extreme Quantile Treatment Effects. *arXiv:2603.23309*.
4. **Li, M.*** & Castro-Camilo, D. (2025). On the importance of tail assumptions in climate extreme event attribution. *arXiv:2507.14019*. Minor revision under review at *Climatic Change*.

Research Software

5. **Li, M.*** & Castro-Camilo, D. (2025). eFCM: An R package for the Exponential Factor Copula Model. CRAN R package. cran.r-project.org/package=eFCM

Research Experience

Causal Structure Learning for Multivariate Extremes

Developed structure learning methods for high-dimensional multivariate extremes and tail dependence, enabling recovery of directional causal structure encoded by directed acyclic graphs in extreme regimes.

Glasgow, UK

2025 – 2026

Causal Inference for Extreme Quantile Treatment Effects

Developed a causal inference framework based on inverse estimating equations for estimating treatment effects in the far tails of outcome distributions, with formal identification and asymptotic guarantees.

Glasgow, UK

2023 – 2025

Tail Dependence and Causal Attribution of Climate Extremes

Integrated multivariate extreme value modeling into a counterfactual causal inference framework to assess the role of tail dependence in extreme event attribution, showing that dependence assumptions critically shape causal conclusions.

Glasgow, UK

2022 – 2025

Modeling Extreme Quantiles and Multivariate Extremes

Co-developed a modeling strategy for extreme quantiles and multivariate extremes, integrating extreme value and dependence modeling techniques. Results published in *Extremes*.

Glasgow, UK

2023 – 2024

Predictive Modeling for Extreme Event Attribution

Glasgow, UK

Motivated by the EVA Data Challenge 2025, developed ConvLSTM-based predictive models for attribution and risk estimation of extreme events, capturing spatio-temporal dependence and tail behaviour; now an independent methodological study with manuscript in preparation.

2025 – 2026

Teaching Experience

University of Glasgow

Glasgow, UK

Tutor and Demonstrator

2022 – 2026

Statistics & Modeling	Bayesian Statistics, Time Series, Flexible Regression, Biostatistics, Advanced Predictive Models
Statistical Computing	Introduction to R & Python, Data Analysis
Project Support	MSc Project (Statistics)
Responsibilities	Led tutorials and lab sessions for undergraduate and MSc students; provided technical support for programming assignments; assessed coursework and provided constructive feedback.

Talks and Presentations

1. **Statistics Seminar, School of Mathematics and Statistics, University of Glasgow**, 23 September 2026, Glasgow (UK). *Talk presentation.*
2. **First workshop of the Glasgow-Edinburgh Extremes Network**, 11 December 2025, University of Glasgow, Glasgow (UK). *Talk presentation.*
3. **ISI World Statistics Congress 2025**, 5–9 October 2025, International Statistical Institute, The Hague (Netherlands). *Talk presentation.*
4. **RSS 2025 International Conference**, 1–4 September 2025, Royal Statistical Society, Edinburgh (UK). *Poster presentation.*
5. **The 6th International Conference on Advances in Extreme Value Analysis and Application to Natural Hazards (EVAN 2024)**, 16–19 July 2024, Istituto Veneto di Scienze, Venice (Italy). *Talk presentation.*
6. **STOR-i Extremes Workshop**, 20–22 September 2023, Lancaster University, Lancaster (UK). *Talk presentation.*
7. **Research Students’ Conference in Probability and Statistics**, 11–14 September 2023, University of Sheffield, Sheffield (UK). *Talk presentation; Travel grant recipient.*

Awards and Recognitions

- **CDSAI Training and Conference Fund (Competitive Award).** University of Glasgow, 2026.
- **Google Cloud Research Grant.** December 2025.
- **3rd Place, IASC Data Analysis Competition, Travel grant awarded.** ISI World Statistics Congress, The Hague (Netherlands), July 2025.
- **Fully funded PhD scholarship.** 2022–2026.
- **Academic Scholarships.** Southwestern University of Finance and Economics. 2015–2019.

Research Leadership & Academic Service

Journal Reviewing

2025 – Present

Journal reviewer: *Data-Centric Engineering*, *Journal of Flood Risk Management*; Co-reviewer: *Annals of Applied Statistics* (with supervisor).

Seminar & Reading Group Organisation

2025 – Present

Co-organized the monthly **Glasgow–Edinburgh Extremes Network (GLE²N)** seminar series, coordinating invited speakers and supporting seminar delivery for sessions hosted at the University of Glasgow. Initiated and organised the thematic session series “Causality for Extremes”.

Summer School Organisation

2026

Co-organizer of the **Edinburgh Summer School on Generative AI for Extremes**, a GLE²N summer school for PhD students and early-career researchers at the interface of generative AI and extreme value theory.

Skills

Programming & Computing	R (advanced), Python, C++, Linux, Git, HPC, Cloud Computing (Google Cloud)
Scientific Software & Reproducibility	R package development (CRAN), R Markdown, LaTeX
Web & Documentation	HTML/CSS, Hugo